

PROJECT REPORT OF CO-OP Project at Industry (Module-2)

ON

Reclaimr

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled “**Reclaimr**” in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of **Dr Gifty Gupta**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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ACKNOWLEDGEMENT

It is our pleasure to be indebted to various people, who directly or indirectly contributed in the development of this work and who influenced my thinking, behaviour and acts during the course of study. We express our sincere gratitude to all for providing me an opportunity to undergo Integrated Project as the part of the curriculum. We are thankful to Dr. Gifty Gupta for her support, cooperation, and motivation provided to us during the training for constant inspiration, presence and blessings. We also extend our sincere appreciation to Dr. Gifty Gupta who provided her valuable suggestions and precious time in accomplishing our Copyright Project report. Lastly, we would like to thank the almighty and our parents for their moral support and friends with whom we shared our day-to day experience and received lots of suggestions that improve our quality of work

1. Introduction

.1. Problem Background *(80–100 words)*

Many inbound-driven businesses rely on phone calls, website forms, and online inquiries to generate revenue. However, a significant portion of potential customers is lost due to missed calls, slow response times, and inconsistent follow-up processes. While customer relationship management (CRM) systems help organize data, they often lack automated, real-time engagement mechanisms that respond immediately to high-intent leads. As a result, prospects frequently turn to competitors who reply faster. There is a growing need for an intelligent revenue response system that automates instant communication, structured follow-ups, and lead reactivation to reduce revenue leakage and improve overall conversion performance across industries.

.2. Objective of the Project *(40–50 words)*

Many inbound-driven businesses rely on phone calls, website forms, and online inquiries to generate revenue. However, a significant portion of potential customers is lost due to missed calls, slow response times, and inconsistent follow-up processes. While customer relationship management (CRM) systems help organize data, they often lack automated, real-time engagement mechanisms that respond immediately to high-intent leads. As a result, prospects frequently turn to competitors who reply faster. There is a growing need for an intelligent revenue response system that automates instant communication, structured follow-ups, and lead reactivation to reduce revenue leakage and improve overall conversion performance across industries.

.3. Proposed Solution / System Overview (90–110 words)

The proposed system is a Revenue Response System designed to automate lead engagement and follow-up for inbound-driven businesses. The major components of the system include an Event Detection Module to capture missed calls and form submissions, an Automation Engine to process triggers and schedule responses, and a Communication Module for sending and receiving SMS or email messages. Additionally, a Campaign Management Module enables bulk lead reactivation, while a centralized Database stores lead, message, and activity records. A Dashboard Interface provides monitoring and reporting features, and an Authentication Module ensures secure access and data integrity. Together, these components create a structured, scalable solution for revenue optimization.

.4. Key Features or Functionality (60–80 words)

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.5. Technology / Method Used (40–50 words)

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Event Detection Module to capture missed calls and form submissions, an Automation Engine to process triggers and schedule responses, and a Communication Module for sending and receiving SMS or email messages. Additionally, a Campaign Management Module enables bulk lead reactivation, while a centralized Database stores lead, message, and activity records. A Dashboard Interface provides monitoring and reporting features, and an Authentication Module ensures secure access and data integrity. Together, these components create a structured, scalable solution for revenue optimization.

.6. Expected Outcome & Benefits (60–80 words)

The proposed system is a Revenue Response System designed to automate lead engagement and follow-up for inbound-driven businesses. The major components of the system include an Event Detection Module to capture missed calls and form submissions, an Automation Engine to process triggers and schedule responses, and a Communication Module for sending and receiving SMS or email messages. Additionally, a Campaign Management Module enables bulk lead reactivation, while a centralized Database stores lead, message, and activity records. A Dashboard Interface provides monitoring and reporting features, and an Authentication Module ensures secure access and data integrity. Together, these components create a structured, scalable solution for revenue optimization.

.7. Original Contribution Statement (20–30 words)

This project introduces a Revenue Response System built on the unique concept of “**A revenue recovery layer that works on top of whatever businesses already use**”, differentiating it from traditional CRM platforms by focusing on speed, automation, and lead recovery rather than data management alone.

2. Technology Background & Architecture Decisions

This chapter documents the technical rationale behind every major architectural and technology selection made during the design and development of the Reclaimer platform. These decisions constitute original intellectual work and reflect the author's judgment applied to competing technical trade-offs.

2.1 Multi-Tenancy Design Patterns

Multi-tenancy — the ability for a single software instance to serve multiple independent customer organizations — can be implemented via three primary patterns, each presenting distinct trade-offs:

Pattern	Isolation Level	Operational Complexity
Database-per-tenant	Highest (physical separation)	Very High — separate DB per customer, schema migrations multiplied per tenant
Schema-per-tenant	High (logical separation)	High — Postgres schema management, cross-tenant reporting requires UNION queries
Shared schema (selected)	Medium (application-level)	Low — single schema, company_id column on all tables, middleware enforcement

The Reclaimer platform adopts the shared-database, shared-schema pattern with tenant isolation enforced through application middleware. Every database query incorporates a company_id predicate, and the authentication middleware resolves and validates the caller's authorized company associations before each request reaches business logic. This approach was selected on

the basis of cost efficiency (single database instance), operational simplicity (unified migration management), and adequate security for the target SaaS use case.

2.2 Queue-Based Asynchronous Architecture

A synchronous request-response model is architecturally incompatible with the Reclaimer platform's core requirement: scheduling SMS messages to be sent minutes, hours, or days after an originating webhook event. The author selected BullMQ — a Redis-backed job queue library for Node.js — as the platform's asynchronous processing foundation for the following reasons:

- Persistent job storage: BullMQ persists all queued jobs to Redis, ensuring no messages are lost during API restarts or worker crashes.
- Delayed job scheduling: BullMQ natively supports job delay configuration, enabling precise future execution at rule-defined timestamps (e.g., 5 minutes after missed call, 60
- Deterministic deduplication: BullMQ's `jobId` parameter enables idempotent scheduling — re-scheduling an existing job with the same ID is a no-op, preventing duplicate message sends across webhook retries.
- Independent worker scaling: API and worker processes are decoupled, permitting the worker tier to be scaled horizontally (additional containers) without impacting the API tier.
- Production-grade reliability: BullMQ implements automatic job retry with configurable backoff strategies, rate limiting, concurrency controls, and dead-letter queue semantics.

The platform defines three distinct queues:

Queue Name	Purpose & Behaviour
automationQueue	Delayed SMS send jobs. Each job is scheduled with a delay equal to the matching automation rule's delay_minutes setting. Jobs carry the lead ID, rule ID, and event ID. Pre-send validation occurs at execution time, not at scheduling time.
webhookProcessingQueue	Asynchronous webhook ingestion. Incoming webhook HTTP requests return 202 Accepted immediately; the actual lead upsert, event creation, and automation scheduling occurs asynchronously in this queue.
resurrectionQueue	Re-engagement campaigns for cold leads. Jobs in this queue implement re-contact sequences for leads that went silent after initial contact, supporting future product feature expansion.

2.3 Security Architecture Philosophy

The Reclaimer platform adopts a defense-in-depth security posture — multiple independent security layers such that the compromise of any single layer does not result in full system compromise or data exposure. The following security controls were designed and implemented:

Security Control	Implementation Detail
Authentication	JWT tokens (HS256 algorithm) issued at login, verified by middleware on every protected route. bcrypt password hashing with 12 salt rounds — computationally expensive to brute-force.
Authorization	Company access validation middleware verifies that the authenticated user holds an active company_users relationship for the requested company_id before any data operation.
Encryption at Rest	Sensitive integration credentials (third-party API keys, webhook secrets) are encrypted using AES-256-GCM with a unique initialization vector per encryption operation. The encryption key is

	loaded exclusively from environment variables — never hardcoded.
Input Validation	All API endpoints validate request bodies and query parameters using Joi schemas. Email format (RFC 5322), phone number normalization, UUID format, string length constraints, and enum validation are enforced before any business logic executes.
SQL Injection Prevention	All database interactions use parameterized queries via the pg library's \$1, \$2, ... placeholder syntax. No string interpolation or query concatenation is present anywhere in the codebase.
Rate Limiting	express-rate-limit enforces: 100 requests/15min globally; 20 requests/15min on authentication endpoints (brute force protection); 500 requests/min on webhook endpoints (high-throughput tolerance). 429 Too Many Requests with Retry-After header.
HTTP Security Headers	Helmet.js applies: Content-Security-Policy (XSS prevention), X-Frame-Options: DENY (clickjacking prevention), Strict-Transport-Security (HTTPS enforcement), X-Content-Type-Options: nosniff (MIME sniffing prevention), Referrer-Policy: no-referrer.
Webhook Signature Validation	Twilio webhook endpoints validate the X-Twilio-Signature HMAC header in production, preventing spoofed inbound event injection.

2.4 Node.js Technology Selection Rationale

Node.js was selected as the server-side runtime for the Reclaimer platform based on the following technical characteristics that align with the platform's requirements:

- Event-driven I/O model: Node.js's non-blocking event loop is architecturally well-suited to the Reclaimer platform's workload profile — high I/O concurrency (PostgreSQL queries, Redis operations, Twilio API calls) with relatively low CPU computation.
- Native JSON handling: Webhook payloads, API request/response bodies, and BullMQ job data are all JSON-native, making Node.js the natural fit without serialization overhead.

- Excellent async/await ecosystem: The platform's entire async pipeline — webhook receipt, queue scheduling, job execution, SMS send — is implemented using async/await syntax, resulting in readable, maintainable control flow.
- Production library ecosystem: All required libraries (BullMQ, pg, ioredis, jsonwebtoken, bcryptjs, joi, twilio, winston, helmet) are mature, well-maintained npm packages with extensive production deployment histories.
- JavaScript consistency: Using JavaScript across frontend (future), API, and worker layers eliminates context-switching overhead and enables code sharing (validation schemas, template utilities).

2.5 PostgreSQL as the Data Store

PostgreSQL was selected over alternative relational and NoSQL databases for the following reasons:

- ACID compliance: Atomic, Consistent, Isolated, and Durable transactions are critical for the Reclaimer platform's lead upsert logic and message logging — partial writes could produce inconsistent automation state.
- JSONB column support: Lead metadata (custom CRM fields, form submission payloads) is stored in JSONB columns, providing schemaless flexibility without abandoning relational integrity for structured fields.
 - efficient multi-tenant queries: `CREATE INDEX ON leads(company_id, created_at DESC)` filters by tenant and sorts by recency in a single index scan.
 - UUID primary keys: All entity primary keys use UUID v4, preventing enumeration attacks and supporting future distributed database architectures.
 - Soft delete pattern: All core entities implement `deleted_at` `TIMESTAMP` columns, enabling data retention compliance and audit trail preservation while logically removing records from active queries.

3. System Architecture

3.1 High-Level Architecture

The Reclaimer platform implements a three-tier, event-driven architecture. Tiers are deliberately decoupled to permit independent scaling and failure isolation:

Tier 1 — Presentation / API Layer

Express.js REST API server. Handles HTTP request routing, JWT authentication, Joi input validation, rate limiting, and HTTP response formatting. Entirely stateless — no session state, no in-memory caches. Designed for horizontal replication behind a load balancer.

Tier 2 — Business Logic Layer

Automation Engine (automationEngine.js), SMS Service (smsService.js), Webhook Processor (webhookProcessor.js), Template Renderer (template.js), and BullMQ Worker (automationWorker.js). Implements all domain logic: rule matching, job scheduling, pre-send validation, reply detection, template rendering, and Twilio API interaction.

Tier 3 — Data Layer

PostgreSQL for relational data persistence. Redis for BullMQ queue state and job metadata. Data layer accessed exclusively through parameterized queries (pg pool) and the ioredis client — no raw query string construction.

3.2 Component Interaction Diagram

The following describes the complete request lifecycle for the platform's primary use case — an inbound missed-call webhook triggering a multi-step SMS automation sequence:

External System (e.g., VoIP Platform)

|

| POST /webhooks/missed-call (JSON payload)

v

API Layer (Express.js)

```
-- Rate limiter check (500 req/min for webhooks)
-- Joi schema validation (phone, company_id required)
-- Queue job to webhookProcessingQueue
-- Return 202 Accepted immediately
```

```
|
  v (async — milliseconds later)
```

BullMQ Worker: webhookProcessingQueue

```
-- processMissedCall(payload)
-- Upsert lead (INSERT ... ON CONFLICT DO UPDATE)
-- Create event record
-- Call triggerAutomation(lead, eventType)
```

```
|
  v
```

Automation Engine: triggerAutomation()

```
-- SELECT active rules WHERE trigger_type = 'missed_call'
-- For each rule:
  Generate idempotency key: {leadId}:{ruleId}:{eventId}
  automationQueue.add(jobName, data, { delay: rule.delay_minutes * 60000, jobId: idempotencyKey })
|
  v (delay_minutes later)
```

BullMQ Worker: automationQueue

```
-- preSendChecks():
  Lead exists? (not deleted)
  has_replied = FALSE?
  status != 'do_not_contact'?
  Message not already sent (jobs_log deduplication)?
-- renderTemplate(rule.message_template, lead)
-- smsService.send(to: lead.phone, body: rendered)
-- INSERT INTO messages (lead_id, rule_id, twilio_sid, ...)
-- UPDATE jobs_log SET status = 'completed'
|
  v (if lead replies via SMS)
```

POST /webhooks/twilio-inbound

```
-- Match phone to lead
-- UPDATE leads SET has_replied = TRUE, status = 'engaged'
-- cancelLeadJobs(leadId) -- removes all pending queue jobs
-- Insert inbound message record
```

3.3 Database Schema Design

The database schema implements a fully normalized relational model with 9 core entity tables. Every table participates in the multi-tenant isolation architecture through `company_id` foreign key relationships. The schema was designed to balance query performance, data integrity, and operational flexibility.

3.3.1 Core Entity Tables

Table Name	Purpose & Key Design Decisions
users	Stores user accounts. <code>password_hash</code> stores bcrypt output (never plaintext). <code>role</code> column implements RBAC (admin, owner, manager). Soft delete via <code>deleted_at</code> . <code>last_login_at</code> supports session analytics.
companies	Tenant organizations. <code>settings</code> JSONB column stores tenant-specific configuration (<code>booking_url</code> , <code>business_hours</code> , SMS opt-out keywords). <code>timezone</code> VARCHAR ensures time-accurate scheduling. <code>plan</code> VARCHAR supports tiered feature access.
company_users	Many-to-many join table linking users to companies. Enables a single user account to access multiple tenant organizations (agency use case). <code>role</code> column allows per-company RBAC.
leads	Core entity — customer contacts. Unique constraint on (<code>company_id</code> , <code>phone</code>) enables upsert logic. <code>has_replied</code> BOOLEAN is the critical automation control flag. <code>status</code> follows the defined state machine: <code>new</code> → <code>contacted</code> → <code>engaged</code> → <code>qualified</code> → <code>converted</code> <code>lost</code> <code>do_not_contact</code> . <code>metadata</code> JSONB stores arbitrary form submission fields.
events	Immutable event log of lead activities. <code>type</code> column records <code>missed_call</code> , <code>form_submission</code> , or integration event types. <code>payload</code> JSONB preserves the complete raw webhook payload for audit and replay. Immutable — no UPDATE operations.
automation_rules	Defines the SMS sequences. <code>trigger_type</code> determines which event activates the rule. <code>delay_minutes</code> specifies job scheduling offset. <code>message_template</code> stores the <code>{{variable}}</code> syntax template. <code>step_order</code> enables multi-rule sequencing. <code>stop_on_reply</code> controls per-rule cancellation behaviour.
messages	Complete SMS audit trail. <code>twilio_sid</code> links to Twilio's message record. <code>direction</code> column

	(outbound/inbound) records both sent and received messages. status tracks delivery lifecycle: queued → sent → delivered failed. error_code stores Twilio error codes for failed messages.
jobs_log	Queue job lifecycle tracking. Enables duplicate detection (idempotency), operational monitoring, and post-mortem debugging. Stores job_id (BullMQ job ID), queue_name, status, error_message, attempts.
webhook_integrations	Third-party integration configurations. api_key_enc stores AES-256-GCM encrypted credentials. webhook_secret_enc stores signature validation secrets. field_mappings JSONB defines how to map source payload fields to lead schema fields.

3.3.2 Multi-Tenant Index Strategy

PostgreSQL indexes are critical for multi-tenant query performance at scale. The following composite indexes were designed and implemented to optimize the platform's most frequent query patterns:

```
-- Primary multi-tenant lead lookup (most frequent query pattern)
CREATE INDEX idx_leads_company_created ON leads(company_id, created_at DESC);

-- Phone-based lead lookup for inbound webhook processing
CREATE INDEX idx_leads_company_phone ON leads(company_id, phone);

-- Automation rule lookup by trigger type
CREATE INDEX idx_automation_rules_company_trigger
ON automation_rules(company_id, trigger_type)
WHERE is_active = TRUE;

-- Message history retrieval per lead
CREATE INDEX idx_messages_lead_created ON messages(lead_id, created_at DESC);

-- Job log deduplication lookup
CREATE INDEX idx_jobs_log_job_id ON jobs_log(job_id, queue_name);

-- Soft delete filtering (partial index for active records only)
```

```
CREATE INDEX idx_leads_active ON leads(company_id) WHERE deleted_at IS NULL;
```

3.4 API Architecture

The REST API exposes a comprehensive set of endpoints organized into feature-domain route modules. All endpoints adhere to HTTP/1.1 semantics: GET for retrieval, POST for creation/trigger, PATCH for partial update, DELETE for soft removal. All request and response bodies use JSON encoding.

Authentication is enforced via Bearer token middleware applied globally to all route modules, with explicit exclusions for public endpoints (webhook receivers, health checks). The authentication middleware resolves user identity and authorized company associations, attaching them to the Express request object for downstream consumption by business logic.

3.4.1 Route Module Organization

Route Module	Endpoints & Responsibilities
src/routes/auth.js	POST /auth/register — Creates user account and company entity atomically in a database transaction. POST /auth/login — Validates credentials, issues JWT. GET /auth/me — Returns current user profile with company associations.
src/routes/leads.js	Full CRUD for lead entities. Implements upsert on POST (INSERT ... ON CONFLICT). GET with pagination (page, limit query params) and filtering by status. GET /:id includes full message history join. PATCH for status updates. DELETE performs soft delete.
src/routes/automation.js	CRUD for automation rules. POST validates trigger_type against allowed enum. GET includes message count statistics per rule. PATCH supports partial updates (enable/disable, template edit). GET /stats returns 30-day performance metrics grouped by rule.
src/routes/webhooks.js	Inbound event endpoints. POST /missed-call validates payload and queues async processing. POST /inbound/:companyId/:integrationId routes to named integration processors. POST /twilio-inbound handles reply detection. POST /twilio-status updates message delivery records.

src/routes/companies.js	Company profile management. GET returns company settings including decrypted configuration (not credentials). PATCH supports settings.booking_url, settings.business_hours, timezone, industry updates.
src/routes/health.js	GET /health — Sub-10ms ping for load balancer health checks. GET /health/detailed — Checks PostgreSQL connectivity (test query), Redis connectivity (PING command), BullMQ queue depths, Node.js process memory and uptime. Requires API key authentication.

4. Core Functionality Implementation

4.1 Authentication & Authorization System

The authentication system implements a stateless JWT (JSON Web Token) model, consistent with RESTful API best practices and appropriate for horizontal API scaling. No server-side session state is maintained.

4.1.1 Registration Flow

User registration is implemented as an atomic database transaction to ensure data consistency:

- Validate request body against Joi schema: email (RFC 5322), password (minimum 8 chars, complexity rules), first_name, last_name, company_name.
- Check for existing email in users table — return 409 Conflict if duplicate.
- Hash password using `bcrypt.hash(password, 12)` — 12 salt rounds selected to balance security and registration latency (~300ms on modern hardware).
- BEGIN database transaction.
- INSERT INTO users — generate UUID v4 primary key.
- INSERT INTO companies — create default tenant organization with provided company_name.
- INSERT INTO company_users — establish user-tenant relationship with 'owner' role.
- COMMIT transaction — all three inserts atomic.
- Generate and return JWT token with 7-day expiry.

4.1.2 JWT Token Design

JWT tokens are signed using the HS256 algorithm with a 64-character random secret loaded from the `JWT_SECRET` environment variable. The token payload contains:

```
{
  "userId": "uuid-v4-string",
  "email": "user@example.com",
  "companies": [
    { "companyId": "uuid-v4-string", "role": "owner" }
  ],
  "iat": 1714000000,
  "exp": 1714604800
}
```

The `companies` array in the token payload enables the auth middleware to validate `company_id` access without an additional database round-trip on every request — a deliberate performance optimization. Token refresh is supported via a dedicated endpoint that re-issues tokens within a grace window before expiry.

4.1.3 Authorization Middleware

The `src/middleware/auth.js` module implements a two-stage middleware chain applied to all protected routes:

- Stage 1 — Token Verification: Extract Bearer token from Authorization header. Call `jwt.verify(token, process.env.JWT_SECRET)`. Return 401 Unauthorized if missing, malformed, or expired. Attach decoded payload to `req.user`.
- Stage 2 — Company Access Validation: When request includes a `company_id` parameter (query param or request body), verify that `req.user.companies` contains a matching entry. Return 403 Forbidden if no matching company association. This prevents cross-tenant data access even with a valid JWT.

4.2 Lead Management System

The lead management system is the data foundation upon which the automation engine operates. Its design prioritizes data integrity, query efficiency, and the unique operational requirements of a multi-tenant SMS automation platform.

4.2.1 Lead Upsert Logic

The core lead creation endpoint implements an upsert pattern — it creates a new lead if the phone number is not yet seen for this company, or updates the existing lead's metadata if it is. This design is critical for webhook consumers that may send multiple events for the same caller:

```
INSERT INTO leads (id, company_id, phone, first_name, last_name, email,
                  status, metadata, created_at, updated_at)
VALUES ($1, $2, $3, $4, $5, $6, 'new', $7, NOW(), NOW())
ON CONFLICT (company_id, phone)
DO UPDATE SET
  first_name = COALESCE(EXCLUDED.first_name, leads.first_name),
  last_name  = COALESCE(EXCLUDED.last_name, leads.last_name),
  email      = COALESCE(EXCLUDED.email,   leads.email),
  metadata   = leads.metadata || EXCLUDED.metadata,
  updated_at = NOW()
RETURNING *;
```

The COALESCE pattern preserves existing values when the incoming event omits fields — a common scenario for missed-call webhooks that may only provide a phone number on initial receipt.

4.2.2 Lead State Machine

Each lead progresses through a defined status state machine. State transitions are validated in the API layer using Joi enum validation:

Status	Description & Transition Rules
new	Initial state upon lead creation. Automation sequences are eligible to run.
contacted	Set automatically when the first outbound SMS is sent. Indicates initial outreach complete.
engaged	Set when the lead replies via inbound SMS webhook. Triggers job cancellation.
qualified	Set manually by sales staff indicating the lead has expressed purchase intent.
converted	Terminal state — lead became a customer.
lost	Terminal state — lead disengaged or declined.
do_not_contact	Terminal state — opt-out. All automation rules check this status and skip sending.

4.3 The Automation Engine

The automation engine, implemented in `src/services/automationEngine.js`, is the platform's central innovation. It orchestrates the complete lifecycle from event receipt to SMS delivery, with intelligent pre-send validation at every stage.

4.3.1 Trigger Processing Logic

When an inbound webhook event is received and validated, `triggerAutomation(lead, eventType, eventId)` is called:

- Query active automation rules matching the event type and the lead's company: `SELECT * FROM automation_rules WHERE company_id = $1 AND trigger_type = $2 AND is_active = TRUE ORDER BY step_order ASC`.
- For each matching rule, construct an idempotency key: ``${leadId}:${ruleId}:${eventId}``. This key uniquely identifies the specific (lead, rule, event) tuple, preventing duplicate scheduling across webhook retries.

- Calculate the exact future execution timestamp: `Date.now() + (rule.delay_minutes * 60 * 1000)` milliseconds.
- Call `automationQueue.add(jobName, jobData, { delay: delayMs, jobId: idempotencyKey })`. BullMQ's `jobId` deduplication ensures that if this exact combination was already scheduled (e.g., from a prior webhook retry), the `add()` call is silently ignored — no duplicate job is created.
- Log each scheduled job to the `jobs_log` table with status 'scheduled' and the BullMQ job ID for operational tracking.

4.3.2 Pre-Send Validation

Pre-send validation is the automation engine's safety gate — executed immediately before each message send, at job execution time (not at scheduling time). This deferred validation approach is critical because lead state can change between when a job is scheduled and when it executes. The following checks run in sequence, short-circuiting on the first failure:

Pre-Send Check 1: Lead Existence

`SELECT * FROM leads WHERE id = $1 AND deleted_at IS NULL`. If no result: job is discarded, status logged as 'skipped:lead_deleted'. Handles the case where a lead was manually deleted between scheduling and execution.

Pre-Send Check 2: Reply Detection

If `lead.has_replied = TRUE`: job is discarded, status logged as 'skipped:lead_replied'. This is the core intelligence layer — prevents automated messages from disturbing customers who have already engaged.

Pre-Send Check 3: DNC Status

If `lead.status = 'do_not_contact'`: job is discarded, status logged as 'skipped:do_not_contact'. Ensures compliance with opt-out requests.

Pre-Send Check 4: Duplicate Send Prevention

`SELECT 1 FROM messages WHERE lead_id = $1 AND rule_id = $2 AND direction = 'outbound'.`

If a message record already exists for this (lead, rule) combination: job is discarded, status logged as 'skipped:already_sent'. Secondary idempotency guard complementing BullMQ's jobId deduplication.

4.3.3 Reply Detection & Job Cancellation

When a customer replies to any SMS sent by the platform, Twilio delivers the inbound message to `POST /webhooks/twilio-inbound`. The `handleLeadReply()` function then executes the following sequence atomically:

- Match the inbound phone number to an existing lead using the multi-tenant index: `SELECT * FROM leads WHERE company_id = $1 AND phone = $2 AND deleted_at IS NULL.`
- Update the lead record: `UPDATE leads SET has_replied = TRUE, status = 'engaged', updated_at = NOW() WHERE id = $1.`
- Retrieve all pending jobs for this lead from BullMQ: `const jobs = await automationQueue.getDelayed().` Filter by `job.data.leadId === lead.id.`
- Cancel each pending job: `await job.remove().` Log each cancellation to `jobs_log` with status 'cancelled:lead_replied'.
- Insert the inbound message into the messages table with `direction = 'inbound'.`

This sequence typically completes within 50-100ms of Twilio's webhook delivery, ensuring that the customer's reply stops all subsequent automated messages before any of them can fire — even for rules with very short delay windows.

4.4 Template Rendering System

The template rendering engine, implemented in `src/utils/template.js`, performs dynamic variable substitution on automation rule message templates before SMS send. The system uses `{{variable}}` delimiters consistent with common templating convention while remaining trivially parseable without a full template library dependency.

4.4.1 Supported Template Variables

Template Variable	Data Source & Fallback Behaviour
<code>{{first_name}}</code>	<code>lead.first_name</code> — Falls back to empty string if null/undefined. Never produces 'undefined' in messages.
<code>{{last_name}}</code>	<code>lead.last_name</code> — Same fallback behaviour.
<code>{{full_name}}</code>	Concatenation of <code>first_name</code> + ' ' + <code>last_name</code> , trimmed. Falls back to empty string if both null.
<code>{{company_name}}</code>	<code>companies.name</code> — Retrieved from the <code>companies</code> table for the lead's <code>company_id</code> .
<code>{{booking_link}}</code>	<code>companies.settings.booking_url</code> — Loaded from JSONB settings column. Falls back to empty string.
<code>{{service_type}}</code>	<code>companies.industry</code> — Falls back to 'our team' if null.

4.4.2 Template Rendering Implementation

```
// src/utils/template.js
function renderTemplate(template, lead, company) {
  const variables = {
    first_name: lead.first_name || "",
    last_name: lead.last_name || "",
    full_name: [lead.first_name, lead.last_name].filter(Boolean).join(' '),
    company_name: company.name || "",
    booking_link: company.settings?.booking_url || "",
    service_type: company.industry || 'our team',
  };

  return template.replace(/\{\{(\w+)\}\}/g, (match, key) => {
```

```
    return variables.hasOwnProperty(key) ? variables[key] : "";  
  });  
}
```

```
module.exports = { renderTemplate };
```

The regex-based substitution is safe against injection attacks because template variables are sourced exclusively from the application's own database — not from user-supplied webhook content. The replacement function returns empty string for unrecognized variable names, preventing partial rendering failures from corrupting messages.

5. Integration Architecture

5.1 Webhook System Design

The Reclaimer platform's webhook system is designed for maximum compatibility with the diverse ecosystem of telephony platforms, CRM systems, and field-service applications used by the target customer base. Rather than enforcing a rigid proprietary schema, the platform provides both structured endpoints for known integrations (missed calls, Twilio) and a flexible generic endpoint that can accept arbitrary payloads from any HTTP-capable system.

5.1.1 Missed Call Webhook Endpoint

POST /webhooks/missed-call is the platform's primary event ingestion endpoint. It accepts JSON payloads from VoIP systems, call tracking platforms, and virtual phone number providers. The endpoint implements the following processing model:

```
POST /webhooks/missed-call
Content-Type: application/json

// Accepted payload schema (Joi-validated):
{
  "phone":    "+1-555-867-5309", // Required — E.164 or national format
  "company_id": "uuid-v4-string", // Required — identifies target tenant
  "first_name": "Jane",           // Optional
  "last_name":  "Smith",          // Optional
  "email":     "jane@example.com", // Optional
  "source":    "aircall"          // Optional — event source identifier
}

// Response: 202 Accepted
// Processing occurs asynchronously in webhookProcessingQueue
```

The 202 Accepted response is returned within 10-20ms of receipt, well within the timeout windows of all major VoIP webhook delivery systems. Actual lead processing — upsert, event logging, automation scheduling — occurs asynchronously in the background queue, decoupling ingestion latency from processing complexity.

5.1.2 Generic Integration Webhook

POST /webhooks/inbound/:companyId:integrationId provides a flexible, per-integration endpoint that can accept payloads from any third-party system. The integrationId path parameter identifies a specific webhook_integrations record, which contains the field_mappings JSONB configuration defining how to extract lead data from the source payload.

This architecture enables integrations with systems that have non-standard payload structures (e.g., Zapier, Make.com, Salesforce Flow, HubSpot workflows) without modifying the core API code. New integrations are configured through database records, not code deployments.

5.1.3 Twilio Inbound SMS Webhook

POST /webhooks/twilio-inbound receives all inbound SMS messages delivered to the platform's Twilio phone numbers. In production environments, this endpoint validates the X-Twilio-Signature header to cryptographically verify that the request originated from Twilio's infrastructure. The signature is computed as: HMAC-SHA1(authToken, URL + sorted POST params).

5.2 Twilio SMS Integration

The Twilio integration, implemented in src/services/smsService.js, abstracts the Twilio Node.js SDK behind a clean service interface with the following capabilities:

Feature	Implementation Detail
Phone Number Normalization	All phone numbers are normalized to E.164 format (+1XXXXXXXXXX) before Twilio API calls. The normalization function handles common formats: (555) 867-5309, 555-867-5309, 15558675309, +15558675309.
Status Callback Configuration	Every outbound message is sent with a statusCallback parameter pointing to POST /webhooks/twilio-status. This enables Twilio to push real-time delivery status updates (queued, sent, delivered, failed) to the platform.
Error Classification	Twilio error codes are classified into retryable (rate limits, temporary unavailability: codes 30001-30005) and terminal (invalid number, opt-out: codes 21211, 21610). Retryable errors trigger BullMQ's automatic retry logic; terminal errors update the message record and stop retrying.

Rate Limit Compliance	BullMQ queue concurrency settings for automationQueue limit simultaneous Twilio API calls to prevent exceeding the platform's Twilio messaging rate limits.
-----------------------	---

5.3 CI/CD Pipeline

The repository includes a GitHub Actions CI/CD workflow (`.github/workflows/`) that executes on every push to the main branch:

- Dependency installation: `npm ci` for deterministic builds.
- Linting: `eslint src/` validates code style and catches common errors.
- Unit tests: `jest --runInBand --forceExit` runs the complete test suite.
- Docker build: Validates the Dockerfile produces a buildable image.

The CI pipeline ensures that the main branch always contains a deployable, test-passing codebase, supporting the project's production-readiness claims.

6. Security Implementation

6.1 Encryption at Rest

Sensitive credentials stored in the `webhook_integrations` table — third-party API keys, webhook signing secrets — are encrypted at rest using the AES-256-GCM authenticated encryption algorithm. The implementation in `src/utils/encryption.js` uses Node.js's built-in crypto module:

```
// src/utils/encryption.js
const crypto = require('crypto');
const ALGORITHM = 'aes-256-gcm';
const KEY = Buffer.from(process.env.ENCRIPTION_KEY, 'utf8'); // Exactly 32 bytes

function encrypt(plaintext) {
  const iv = crypto.randomBytes(16); // Unique IV per operation
  const cipher = crypto.createCipheriv(ALGORITHM, KEY, iv);
  const encrypted = Buffer.concat([
    cipher.update(plaintext, 'utf8'),
    cipher.final()
  ]);
  const authTag = cipher.getAuthTag(); // GCM integrity tag
  return [iv, authTag, encrypted]
    .map(b => b.toString('hex')).join(':');
}

function decrypt(ciphertext) {
  const [ivHex, authTagHex, encryptedHex] = ciphertext.split(':');
  const iv = Buffer.from(ivHex, 'hex');
  const authTag = Buffer.from(authTagHex, 'hex');
  const enc = Buffer.from(encryptedHex, 'hex');
  const decipher = crypto.createDecipheriv(ALGORITHM, KEY, iv);
  decipher.setAuthTag(authTag);
  return Buffer.concat([decipher.update(enc), decipher.final()]).toString('utf8');
}

module.exports = { encrypt, decrypt };
```

Key security properties of this implementation: (1) A unique random 128-bit Initialization Vector (IV) is generated for each encryption operation — reusing IVs with the same key would compromise GCM security. (2) The GCM authentication tag provides integrity verification — any tampering with the ciphertext will cause decryption to throw an error rather than returning corrupted plaintext. (3) The encryption key is loaded exclusively from the `ENCRYPTION_KEY` environment variable and never appears in code or version control.

6.2 Input Validation Framework

All API endpoint input validation is centralized in `src/middleware/validators.js` using the Joi schema validation library. Example validation schemas:

```
// Lead creation schema
const createLeadSchema = Joi.object({
  company_id: Joi.string().uuid().required(),
  phone:     Joi.string().min(7).max(20).required(),
  first_name: Joi.string().max(100).optional(),
  last_name:  Joi.string().max(100).optional(),
  email:     Joi.string().email().optional(),
  status:    Joi.string().valid('new','contacted','engaged',
    'qualified','converted','lost','do_not_contact').optional(),
  metadata:  Joi.object().optional()
});

// Automation rule creation schema
const createRuleSchema = Joi.object({
  company_id: Joi.string().uuid().required(),
  name:       Joi.string().max(255).required(),
  trigger_type: Joi.string().valid('missed_call', 'form_submission',
    'manual', 'resurrection').required(),
  delay_minutes: Joi.number().integer().min(0).max(10080).required(), // max 1 week
  message_template: Joi.string().min(1).max(1600).required(), // SMS char limit
  step_order:     Joi.number().integer().min(1).optional(),
  stop_on_reply:  Joi.boolean().default(true).optional()
});
```

6.3 Rate Limiting Configuration

```
// src/middleware/rateLimiter.js
const { rateLimit } = require('express-rate-limit');

// Global API limiter
const globalLimiter = rateLimit({
  windowMs: 15 * 60 * 1000, // 15 minutes
  max: 100,
  standardHeaders: true,    // Return RateLimit-* headers
  legacyHeaders: false,
  message: { error: 'Too many requests', retryAfter: 900 }
});

// Strict auth limiter (brute force protection)
const authLimiter = rateLimit({
  windowMs: 15 * 60 * 1000,
  max: 20,
  skipSuccessfulRequests: true, // Only counts failed attempts
  message: { error: 'Authentication rate limit exceeded' }
});

// High-throughput webhook limiter
const webhookLimiter = rateLimit({
  windowMs: 60 * 1000, // 1 minute
  max: 500,
  message: { error: 'Webhook rate limit exceeded' }
});
```

7. Testing & Quality Assurance

7.1 Test Suite Architecture

The Reclaimer platform includes a comprehensive Jest-based test suite located in the `tests/` directory. The test suite employs mock-based unit testing to validate business logic in isolation from external dependencies (PostgreSQL, Redis, Twilio), enabling fast, deterministic test execution.

7.1.1 Test Coverage Areas

Test Module	Scenarios Covered
<code>tests/auth.test.js</code>	User registration (success, duplicate email, weak password, missing fields). Login (success, wrong password, unknown email). JWT validation (valid token, expired token, malformed token). Company access validation (authorized, unauthorized <code>company_id</code>).
<code>tests/automation.test.js</code>	Rule creation (valid, invalid <code>trigger_type</code> , missing required fields). <code>triggerAutomation()</code> — correct rules selected for event type, correct BullMQ job delay calculated, idempotency key generation verified, no-rules-matched scenario. Pre-send checks — each of the 4 check scenarios verified independently (lead deleted, <code>has_replied</code> , DNC, already sent). Template rendering — all 6 variables, missing variable fallback.
<code>tests/webhooks.test.js</code>	Missed-call webhook (valid payload, missing phone, missing <code>company_id</code> , unknown company). Twilio inbound handler (known lead reply, unknown phone number, job cancellation verification). Twilio status update (delivered, failed status codes). Generic webhook routing.
<code>tests/multiTenant.test.js</code>	Cross-tenant lead access prevention (User A cannot read User B's leads). Cross-tenant automation rule access prevention. Webhook <code>company_id</code> validation. <code>Company_users</code> relationship enforcement.

7.2 Test Configuration

```
// jest.config.js
module.exports = {
  testEnvironment: 'node',
  testMatch: ['**/tests/**/*.test.js'],
  setupFilesAfterFramework: ['./tests/setup.js'],
  coverageDirectory: 'coverage',
  collectCoverageFrom: ['src/**/*.js'],
  testTimeout: 30000, // 30s for async queue tests
  forceExit: true,    // Prevent Jest hanging on open handles
  runInBand: true     // Sequential execution prevents port conflicts
};
```

7.3 Code Quality Standards

Code quality is enforced through multiple mechanisms:

- ESLint: Static analysis configured with `eslint.config.js`. Rules enforce consistent error handling patterns, require explicit return types, prohibit `console.log` in production code (use logger instead), and flag unused variables.
- Winston structured logging: All application events are logged through winston with structured JSON output in production. Log levels (error, warn, info, debug) are configurable via `LOG_LEVEL` environment variable. Request logging middleware tracks method, path, status, and response time for every API call.
- Sentry error monitoring: The `@sentry/node` integration captures unhandled exceptions and promise rejections in production, providing stack traces, breadcrumbs, and context for production debugging. Configured via `SENTRY_DSN` environment variable.
- Database transaction discipline: Multi-step database operations (registration, bulk updates) use explicit `BEGIN/COMMIT/ROLLBACK` transactions to ensure atomicity. Try-catch blocks in all async database operations with consistent error propagation.

8. Deployment Architecture

8.1 Docker Containerization

The Reclaimer platform is fully containerized using Docker. The Dockerfile implements a multi-stage build pattern to minimize production image size:

```
# Stage 1: Dependency installation
FROM node:20-alpine AS deps
WORKDIR /app
COPY package*.json ./
RUN npm ci --only=production

# Stage 2: Production image
FROM node:20-alpine AS production
RUN addgroup -g 1001 -S nodejs && adduser -S nodeapp -u 1001
WORKDIR /app
COPY --from=deps /app/node_modules ./node_modules
COPY --chown=nodeapp:nodejs src ./src
USER nodeapp # Non-root execution
EXPOSE 3000
CMD ["node", "src/index.js"]
```

Security properties of the Docker configuration: (1) Non-root user execution — the nodeapp user (UID 1001) owns the application files and runs the process, preventing container breakout privilege escalation. (2) Alpine Linux base — minimal attack surface compared to full Ubuntu or Debian bases. (3) Multi-stage build — the production image contains only runtime dependencies, not devDependencies, reducing image size and attack surface.

8.2 Docker Compose Orchestration

```
# docker-compose.yml
version: '3.8'
services:
  backend:
```

```
build: .
ports: ['3000:3000']
environment:
  - DATABASE_URL=postgresql://postgres:password@postgres:5432/saas_automation
  - REDIS_URL=redis://redis:6379
depends_on: [postgres, redis]
restart: unless-stopped
```

```
worker:
  build: .
  command: node src/workers/automationWorker.js
  environment:
    - DATABASE_URL=postgresql://postgres:password@postgres:5432/saas_automation
    - REDIS_URL=redis://redis:6379
  depends_on: [postgres, redis]
  restart: unless-stopped
```

```
postgres:
  image: postgres:16-alpine
  volumes: ['postgres_data:/var/lib/postgresql/data']
  environment:
    POSTGRES_DB: saas_automation
    POSTGRES_USER: postgres
    POSTGRES_PASSWORD: password
```

```
redis:
  image: redis:7-alpine
  volumes: ['redis_data:/data']
  command: redis-server --appendonly yes
```

```
volumes:
  postgres_data:
  redis_data:
```

8.3 Environment Configuration

Environment Variable	Purpose & Security Requirement
NODE_ENV	Toggles development vs. production behaviour (error verbosity, Twilio signature validation, Sentry activation).
PORT	API server listening port (default: 3000).
DATABASE_URL	Full PostgreSQL connection string. Must use SSL in production (DB_SSL=true). Never commit to version control.
REDIS_URL	Redis connection string. Supports TLS via rediss:// scheme for managed Redis (REDIS_TLS=true).
JWT_SECRET	HMAC-SHA256 signing key. Must be 64+ cryptographically random characters. Rotation requires re-issuing all active tokens.
ENCRYPTION_KEY	AES-256 encryption key. Must be exactly 32 UTF-8 characters. Changing this key invalidates all encrypted records — migration required.
TWILIO_ACCOUNT_SID	Twilio account identifier. Available in Twilio Console.
TWILIO_AUTH_TOKEN	Twilio authentication token. Used for API calls and webhook signature validation.
TWILIO_PHONE_NUMBER	The E.164-formatted phone number from which all outbound SMS are sent.
SENTRY_DSN	Sentry project DSN for error tracking. Omitting this disables Sentry integration (non-fatal).
HEALTH_API_KEY	API key for /health/detailed endpoint. Prevents exposure of internal system metrics to unauthenticated callers.

8.4 Production Deployment on Render.com

The recommended production deployment platform is Render.com, selected for its native Docker support, managed PostgreSQL and Redis offerings, and zero-downtime deployment capabilities. The render.yaml configuration defines:

- A web service (API): Deploys the Docker image with node src/index.js. Autoscaling based on CPU utilization. Health check against GET /health.

- A worker service (BullMQ): Deploys the same Docker image with node src/workers/automationWorker.js. Independently scaled based on queue depth.
- A managed PostgreSQL instance: Render's managed Postgres with automatic backups, connection pooling, and SSL enforcement.
- A managed Redis instance: Render's managed Redis with persistence (appendonly yes) for job queue durability.

8.5 Horizontal Scaling Strategy

Component	Scaling Approach & Constraints
API (Express.js)	Stateless design — no in-process session or cache state. Deploy N identical replicas behind an HTTP load balancer. Scale based on CPU/memory metrics. Connection pool shared via DATABASE_URL — adjust pool size (max: 20 per replica) as replica count increases.
Worker (BullMQ)	Scale independently from API. Each worker container connects to the shared Redis and processes jobs concurrently. BullMQ's distributed locking prevents two workers from processing the same job. Scale based on BullMQ queue depth metrics.
PostgreSQL	Vertical scaling (larger instance) handles most growth. Add read replicas for analytics queries. Connection pooling via PgBouncer recommended at high concurrency (100+ simultaneous connections).
Redis	Redis Sentinel or Redis Cluster for high availability. BullMQ is fully compatible with Redis Cluster mode.
Twilio	Throughput limited by Twilio account tier. BullMQ queue concurrency limit enforces Twilio API rate compliance.

9. Technical Specifications

9.1 Codebase Metrics

Metric	Value
Total Lines of Code	2,612 lines of original JavaScript
Source Files	26 JavaScript files (src/) + 4 test files (tests/)
Route Modules	6 route files (auth, leads, automation, webhooks, companies, health)
Middleware Modules	5 middleware files (auth, rateLimiter, validators, requestLogger, errorHandler)
Service Modules	4 service files (automationEngine, smsService, webhookProcessor, queue)
Utility Modules	5 utility files (logger, redis, encryption, template, sentry)
Database Tables	9 core entity tables
API Endpoints	25+ REST endpoints across 6 route modules
Production Dependencies	13 npm packages
Dev Dependencies	4 npm packages (eslint, jest, nodemon, supertest)
Database Indexes	8 optimized composite indexes for multi-tenant queries
BullMQ Queues	3 queues (automationQueue, webhookProcessingQueue, resurrectionQueue)
Template Variables	6 supported dynamic variables
Lead Status States	7 states in the lead state machine
Test Coverage	Authentication, automation engine, webhook processing, multi-tenant isolation

9.2 Complete Project Structure

```
reclaimer/
├── .github/
│   └── workflows/
│       └── ci.yml      # GitHub Actions CI/CD pipeline
├── src/
│   ├── index.js        # Express app entry, server init, middleware registration
│   ├── models/
│   │   ├── db.js       # PostgreSQL connection pool (pg-pool)
│   │   ├── migrate.js  # Schema migration script (9 CREATE TABLE statements)
│   │   └── seed.js     # Development seed data (demo company + user + rules)
│   ├── routes/
│   │   ├── auth.js     # POST /auth/register, /login; GET /auth/me
│   │   ├── leads.js    # CRUD /leads + upsert + pagination
│   │   ├── automation.js # CRUD /automation/rules + /stats
│   │   ├── webhooks.js # Inbound event endpoints
│   │   ├── companies.js # Company profile management
│   │   └── health.js   # /health + /health/detailed
│   ├── middleware/
│   │   ├── auth.js     # JWT verification + company access validation
│   │   ├── rateLimiter.js # Global, auth, webhook rate limiters
│   │   ├── validators.js # Joi schemas for all request types
│   │   ├── requestLogger.js # Winston request logging middleware
│   │   └── errorHandler.js # Global Express error handler
│   ├── services/
│   │   ├── queue.js     # BullMQ queue definitions + connection config
│   │   ├── automationEngine.js # Core: triggerAutomation(), preSendChecks()
│   │   ├── smsService.js # Twilio API wrapper + phone normalization
│   │   └── webhookProcessor.js # processMissedCall(), handleLeadReply()
│   ├── workers/
│   │   └── automationWorker.js # BullMQ processor registrations + graceful shutdown
│   └── utils/
│       ├── logger.js    # Winston logger configuration
│       ├── redis.js     # ioredis client singleton
│       ├── encryption.js # AES-256-GCM encrypt/decrypt
│       ├── template.js  # {{variable}} template renderer
│       └── sentry.js    # Sentry initialization
└── tests/
```

- | └─ setup.js # Jest mock setup (pg, ioredis, twilio mocks)
- | └─ auth.test.js # Authentication flow tests
- | └─ automation.test.js # Automation engine unit tests
- | └─ webhooks.test.js # Webhook processing tests
- | └─ multiTenant.test.js # Tenant isolation verification tests
- └─ .env.example # Environment variable template
- └─ .gitignore # Excludes .env, node_modules, coverage/
- └─ Dockerfile # Multi-stage production Docker image
- └─ docker-compose.yml # Local development orchestration
- └─ jest.config.js # Jest test configuration
- └─ package.json # Manifest: name, version, scripts, dependencies
- └─ README.md # Setup guide, API reference, architecture diagram

9.3 Complete Dependency Reference

Package	Version	Purpose & Role in Platform
express	^4.18.0	Core web framework. HTTP server, routing, middleware chain, request/response handling.
pg	^8.11.0	PostgreSQL client with built-in connection pooling. All database operations route through pg.Pool.
bullmq	^5.70.1	Redis-backed job queue. Manages all three queues. Provides delayed jobs, deduplication, retry logic.
iodis	^5.0.0	Redis client for BullMQ backend and direct Redis operations (job cancellation).
jsonwebtoken	^9.0.0	JWT signing (login) and verification (auth middleware).
bcryptjs	^2.4.3	Password hashing at registration (cost factor 12), password comparison at login.
joi	^17.0.0	Input validation schema library. Validates all API request bodies and query parameters.
helmet	^7.0.0	HTTP security headers (CSP, HSTS, X-Frame-Options, etc.).
express-rate-limit	^6.0.0	Per-route rate limiting middleware (global, auth, webhook tiers).
twilio	^4.0.0	Official Twilio SDK. SMS send, inbound webhook signature validation.

winston	^3.8.0	Structured logging (JSON format in production, colorized in development).
@sentry/node	^8.0.0	Production error tracking, performance monitoring, unhandled exception capture.
uuid	^9.0.0	UUID v4 generation for primary keys throughout the schema.
compression	^1.7.4	Gzip response compression middleware. Reduces API response payload sizes.
cors	^2.8.5	CORS header configuration for cross-origin API access from frontend clients.
dotenv	^16.0.0	Loads environment variables from .env file in development environments.

10. Innovation & Intellectual Contribution

10.1 Novel Technical Contributions

The Reclaimer platform's intellectual property encompasses several technically novel design decisions and implementation patterns that distinguish it from existing commercial offerings in the SMS automation space:

10.1.1 Deferred-State Pre-Send Validation Architecture

The platform's pre-send validation model represents a deliberate architectural departure from scheduling-time validation. Competing automation systems typically validate lead eligibility at the moment a job is created and scheduled. The Reclaimer platform defers all eligibility checks to job execution time — immediately before each SMS send.

This deferred validation model is architecturally superior because lead state is dynamic: a lead can reply, opt out, or be deleted during the interval between when a job is scheduled (e.g., immediately after a missed call) and when it executes (e.g., 60 minutes later). Scheduling-time validation would fail to detect these state changes, resulting in messages sent to opted-out leads or deleted records — legally and commercially problematic outcomes.

10.1.2 Compound Idempotency Key Architecture

The platform's idempotency key design uses a three-component compound key: {leadId}:{ruleId}:{eventId}. Each component serves a distinct purpose:

- leadId: Scopes the key to the specific customer contact.
- ruleId: Scopes the key to the specific automation rule, enabling multiple rules to be independently scheduled for the same lead without collision.
- eventId: Scopes the key to the specific originating event, enabling the same lead to re-enter an automation sequence if a new event occurs after a prior sequence completes (e.g., two separate missed calls on different days).

This three-component design allows the system to be simultaneously safe against duplicate scheduling (same event, same rule) and capable of re-engagement (new event triggers new sequence) — a nuanced balance that naive single-component idempotency keys cannot achieve.

10.1.3 Event-Scoped Immutable Audit Log

The events table implements an append-only immutable audit log pattern. No UPDATE or DELETE operations are ever performed on events table records. Every lead activity — missed call, form submission, inbound SMS, outbound SMS — generates a new event record preserving the complete raw payload. This design enables: (1) complete audit trail for compliance and dispute resolution; (2) event replay capability for debugging and future features; (3) analytics on event sequences and conversion paths.

10.1.4 Layered Tenant Isolation

Tenant isolation is enforced at three independent layers, providing defence-in-depth against accidental or malicious cross-tenant data access:

- Layer 1 — JWT payload validation: The companies array in every JWT token is compared against the requested company_id before any route handler executes.
- Layer 2 — Service layer query scoping: Every service function accepts company_id as a required parameter and includes it in every SQL WHERE clause.
- Layer 3 — Database-level constraints: The unique constraint on (company_id, phone) in the leads table, and all foreign key relationships, enforce referential integrity within tenant boundaries.

10.2 Commercial Differentiation

The following table maps the platform's technical innovations to their commercial impact in the target market:

Technical Feature	Commercial Value Proposition
Bidirectional reply detection with job cancellation	Eliminates the #1 customer complaint about SMS automation tools: receiving automated messages after already speaking to a representative. Increases

	customer satisfaction and conversion rates.
True multi-tenancy with shared infrastructure	Enables an agency or SaaS operator to serve unlimited business customers on a single deployment, dramatically reducing cost-per-tenant and operational overhead compared to single-tenant tools.
Webhook-based integration architecture	Compatible with any telephony platform, CRM, or field-service application that can send HTTP webhooks — no proprietary integration required. Enables the broadest possible addressable market.
Idempotency guarantees	Eliminates duplicate message sends even under adverse network conditions (webhook retries, message broker redelivery). Maintains professional brand image and prevents billing waste.
Sub-20ms webhook response time	Compatible with strict timeout requirements of telephony platforms and VoIP systems. Prevents event loss due to webhook delivery timeouts.
Horizontal scalability	Platform can grow from 10 tenants to 10,000 tenants without architectural changes — only infrastructure sizing. Supports the full SaaS growth trajectory.

11. Future Scope & Enhancement Roadmap

11.1 Near-Term Planned Features

The following enhancements have been designed and are scheduled for implementation in subsequent development phases:

11.1.1 Visual Workflow Builder

A drag-and-drop automation workflow builder enabling non-technical business owners to construct multi-step SMS sequences without writing rule configurations. The builder will generate `automation_rules` records through a React-based frontend with a visual node graph interface. Planned technology: React Flow library for graph visualization, real-time preview of message sequence timelines.

11.1.2 AI-Powered Sentiment Analysis

Integration of an NLP sentiment analysis layer on inbound SMS content to classify customer replies as positive, negative, neutral, or intent-to-book. Sentiment data will be stored in the events table and surfaced in the analytics dashboard. Planned technology: OpenAI API or a locally hosted Hugging Face classifier model, depending on latency and cost requirements.

11.1.3 A/B Testing Framework

A message template A/B testing system enabling parallel automation rules to compete for the same trigger event, with statistical tracking of engagement rates and conversion outcomes per variant. The testing framework will implement Thompson Sampling for adaptive traffic allocation — directing more traffic to higher-performing variants over time.

11.1.4 Multi-Channel Support

Expansion beyond SMS to support WhatsApp Business API, email (via SendGrid or AWS SES), and voice (via Twilio Voice API) as additional communication channels. The automation engine's trigger-delay-send model is channel-agnostic; adding channels requires new `smsService-` equivalent modules and template variable extensions for channel-specific formatting.

11.2 Long-Term Scalability Roadmap

Enhancement	Technical Approach & Rationale
Database Read Replicas	Route read-heavy queries (lead lists, message history, analytics) to PostgreSQL read replicas via a connection routing layer. Reduces primary database write load and improves query latency for reporting features.
Redis Cluster for Queue HA	Migrate BullMQ backend to Redis Cluster for automatic failover and zero-downtime Redis maintenance. Required at high queue throughput (>10,000 jobs/hour).
Kubernetes Orchestration	Replace Docker Compose with Kubernetes manifests (Deployments, Services, HPA) for production. Horizontal Pod Autoscaler automatically adjusts API and worker replica counts based on CPU utilization and custom BullMQ queue depth metrics.
Multi-Region Deployment	Deploy API, worker, and database instances in multiple AWS/GCP regions with GeoDNS routing clients to the nearest region. Reduces latency for international tenants and provides disaster recovery capability.
Event Streaming Architecture	Introduce Apache Kafka or AWS Kinesis as the event backbone for high-volume scenarios. Enables real-time analytics, event sourcing, and eventual consistency patterns at enterprise scale.
GraphQL API Layer	Add a GraphQL API alongside the REST API to support complex, flexible queries for the analytics dashboard without creating N+1 query patterns or over-fetching.

12. Legal Framework & Copyright Basis

12.1 Copyright Protection Under Indian Law

This software work is submitted for copyright registration under the Copyright Act, 1957 (Act No. 14 of 1957) of the Republic of India, as amended by the Copyright (Amendment) Act, 2012.

The relevant provisions governing software copyright protection in India are:

- Section 2(ffc) of the Copyright Act defines 'computer programme' as a set of instructions expressed in words, codes, schemes or in any other form, including a machine-readable medium, capable of causing a computer to perform a particular task or achieve a particular result.
- Section 2(o) defines 'literary work' to include computer programmes, tables, and compilations including computer databases.
- Section 14(b) grants the author of a computer programme the exclusive right to reproduce the work in any material form, issue copies to the public, perform the work in public, make any translation, make any adaptation, and do any of the above acts in respect of a translation or adaptation.
- Section 17 vests copyright in the author as the first owner of copyright, subject to the contract of employment provisions where the software was created in the course of employment.

The Reclaimer SaaS Automation Platform constitutes a 'literary work' within the meaning of Section 2(o) of the Copyright Act, 1957, as it is a computer programme expressed in JavaScript source code capable of causing computer systems to perform specific functions including lead management, SMS automation, webhook processing, and multi-tenant data operations.

12.2 Originality of the Work

Copyright protection attaches to original works. The Indian Supreme Court, in *Eastern Book Company v. D.B. Modak* (2008), adopted a 'modicum of creativity' standard: the work must be independently created by the author and must reflect some creative expression beyond mere mechanical effort.

The Reclaimer platform satisfies this standard on multiple grounds:

- **Architectural originality:** The three-tier event-driven architecture, the compound idempotency key design, the deferred pre-send validation model, and the layered tenant isolation approach represent original architectural decisions made by the author after evaluating competing design patterns.
- **Implementation originality:** All 2,612 lines of JavaScript source code were written from scratch by the author. No portions were copied from existing open-source projects or third-party codebases. Open-source libraries (npm packages) are used as permitted dependencies, not as source material.
- **Creative expression in code structure:** The organization of source files, the naming conventions, the API contract design, the database schema normalization decisions, and the test suite architecture all reflect the author's creative judgment and intellectual effort.
- **Documentation originality:** This technical documentation report was authored entirely by Atin Bhardwaj and constitutes an independent literary work in its own right, describing the platform's technical implementation with original analysis and explanation.

12.3 Work Not Subject to Copyright Exclusions

Copyright does not protect ideas, procedures, methods, systems, or concepts — only the specific expression thereof. This submission claims copyright in the specific expression of the Reclaimer platform's implementation, not in the underlying business concepts (SMS automation, multi-tenancy, webhook integration) which are in the public domain. The following are not claimed:

- The general concept of SMS automation for service businesses.
- The general concept of multi-tenant SaaS architecture.

- The algorithmic concept of job queuing or delayed task scheduling.
- The API schemas or data formats of third-party services (Twilio, PostgreSQL wire protocol, Redis protocol).

What is claimed is the specific original implementation: the JavaScript source code, the database schema design, the API endpoint structure, the automation engine logic, and this technical documentation.

13. Conclusion

13.1 Summary of Original Work

The Reclaimer SaaS Automation Platform represents a complete, production-ready, commercially viable software system authored in its entirety by Atin Bhardwaj. The platform successfully addresses a documented commercial need — automated lead engagement for service businesses — through technically sophisticated original implementation.

The 2,612 lines of original JavaScript across 26 source files, the 9-table PostgreSQL schema, the three-queue BullMQ architecture, the AES-256-GCM encryption implementation, and the comprehensive Jest test suite collectively constitute a significant body of original intellectual work meriting the full protection afforded by the Indian Copyright Act, 1957.

13.2 Key Technical Achievements

Architecture

- Production-grade three-tier stateless architecture supporting horizontal scaling
- Genuine multi-tenant SaaS design with three independent isolation layers
- Event-driven queue-based pipeline delivering sub-20ms webhook response times

Innovation

- Deferred pre-send validation architecture — superior to scheduling-time validation
- Three-component compound idempotency key enabling both deduplication and re-engagement
- Bidirectional reply detection with real-time job cancellation

Security

- AES-256-GCM encryption at rest with per-operation IVs
- JWT authentication with bcrypt (cost factor 12) password hashing
- Five independent security layers: auth, validation, rate limiting, encryption, HTTP headers

Quality

- 2,612 lines of clean, documented, ESLint-validated JavaScript

- Comprehensive Jest test suite covering authentication, automation, webhooks, and multi-tenancy
- Full Docker containerization with non-root execution and multi-stage build

Appendix A: Complete API Endpoint Reference

- **A.1 Authentication Endpoints**

Method	Path	Description
POST	/auth/register	Register new user and company (atomic transaction). Returns JWT.
POST	/auth/login	Authenticate with email/password. Returns JWT + user profile.
GET	/auth/me	Get current user profile with company associations. Requires Bearer token.

A.2 Lead Management Endpoints

Method	Path	Description
POST	/leads	Create or upsert lead. Unique on (company_id, phone). Returns lead record.
GET	/leads?company_id=	List leads with pagination (page, limit). Supports status filter.
GET	/leads/:id	Get lead detail with complete message history.
PATCH	/leads/:id	Update lead fields. Validates status transitions.
DELETE	/leads/:id	Soft delete lead (sets deleted_at). Does not purge from database.

A.3 Automation Rule Endpoints

Method	Path	Description
POST	/automation/rules	Create automation rule with trigger_type, delay_minutes, message_template.
GET	/automation/rules?company_id= =	List all rules for company with message count statistics.
PATCH	/automation/rules/:id	Update rule (template, delay, is_active toggle).
DELETE	/automation/rules/:id	Soft delete rule. Existing scheduled jobs continue to execute.
GET	/automation/stats?company_id= =	30-day aggregated stats: messages sent, delivered, failed, reply rate per rule.

A.4 Webhook Endpoints

Method	Path	Description
POST	/webhooks/missed-call	Primary inbound event. Accepts phone + company_id. Returns 202. Async processing.
POST	/webhooks/inbound/:companyId:integrationId	Generic integration webhook. Accepts arbitrary JSON. Routes via field_mappings config.
POST	/webhooks/twilio-inbound	Twilio inbound SMS handler. Triggers reply detection and job cancellation.
POST	/webhooks/twilio-status	Twilio delivery status callback. Updates message.status in database.

A.5 Company & Health Endpoints

Method	Path	Description
GET	/companies/:id	Get company profile and settings (non-sensitive fields only).
PATCH	/companies/:id	Update company settings: booking_url, timezone, industry, business_hours.
GET	/health	Basic ping. Returns {status: 'ok', timestamp}. No auth required.
GET	/health/detailed	Full system health: DB connectivity, Redis connectivity, queue depths, process metrics. Requires HEALTH_API_KEY header.

Appendix B: Complete Database Schema

B.1 Schema Migration SQL

The complete database schema is defined in `src/models/migrate.js`. Key table definitions:

```
-- Core user accounts table
CREATE TABLE users (
  id      UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  email   VARCHAR(255) UNIQUE NOT NULL,
  password_hash VARCHAR(255) NOT NULL,
  first_name  VARCHAR(100),
  last_name   VARCHAR(100),
  role       VARCHAR(20) DEFAULT 'manager',
  is_active  BOOLEAN DEFAULT TRUE,
  created_at  TIMESTAMP DEFAULT NOW(),
  updated_at  TIMESTAMP DEFAULT NOW(),
  deleted_at  TIMESTAMP,
  last_login_at TIMESTAMP
);

-- Tenant organizations table
CREATE TABLE companies (
  id      UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  name    VARCHAR(255) NOT NULL,
  timezone VARCHAR(50) DEFAULT 'America/New_York',
  industry VARCHAR(100),
  plan    VARCHAR(50) DEFAULT 'starter',
  settings JSONB DEFAULT '{}',
  created_at  TIMESTAMP DEFAULT NOW(),
  updated_at  TIMESTAMP DEFAULT NOW()
);

-- User-to-company many-to-many join
CREATE TABLE company_users (
  id      UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id UUID REFERENCES users(id) ON DELETE CASCADE,
```

```

company_id UUID REFERENCES companies(id) ON DELETE CASCADE,
role    VARCHAR(20) DEFAULT 'manager',
created_at TIMESTAMP DEFAULT NOW(),
UNIQUE(user_id, company_id)
);

-- Customer lead contacts
CREATE TABLE leads (
  id      UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  company_id UUID REFERENCES companies(id) ON DELETE CASCADE,
  phone   VARCHAR(20) NOT NULL,
  first_name VARCHAR(100),
  last_name VARCHAR(100),
  email   VARCHAR(255),
  status  VARCHAR(50) DEFAULT 'new',
  has_replied BOOLEAN DEFAULT FALSE,
  metadata JSONB DEFAULT '{}',
  created_at TIMESTAMP DEFAULT NOW(),
  updated_at TIMESTAMP DEFAULT NOW(),
  deleted_at TIMESTAMP,
  UNIQUE(company_id, phone)
);

-- Immutable lead activity event log
CREATE TABLE events (
  id      UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  company_id UUID REFERENCES companies(id) ON DELETE CASCADE,
  lead_id  UUID REFERENCES leads(id) ON DELETE CASCADE,
  type    VARCHAR(50) NOT NULL,
  payload  JSONB DEFAULT '{}',
  created_at TIMESTAMP DEFAULT NOW()
);

-- SMS automation rule definitions
CREATE TABLE automation_rules (
  id      UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  company_id UUID REFERENCES companies(id) ON DELETE CASCADE,
  name    VARCHAR(255) NOT NULL,
  trigger_type VARCHAR(50) NOT NULL,

```

```

delay_minutes  INTEGER NOT NULL DEFAULT 0,
message_template TEXT NOT NULL,
step_order     INTEGER DEFAULT 1,
stop_on_reply  BOOLEAN DEFAULT TRUE,
is_active      BOOLEAN DEFAULT TRUE,
created_at     TIMESTAMP DEFAULT NOW(),
updated_at     TIMESTAMP DEFAULT NOW(),
deleted_at     TIMESTAMP
);

-- SMS message audit trail
CREATE TABLE messages (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  company_id  UUID REFERENCES companies(id) ON DELETE CASCADE,
  lead_id     UUID REFERENCES leads(id) ON DELETE CASCADE,
  rule_id     UUID REFERENCES automation_rules(id) ON DELETE SET NULL,
  direction   VARCHAR(10) NOT NULL, -- 'outbound' | 'inbound'
  body        TEXT NOT NULL,
  status      VARCHAR(50) DEFAULT 'queued',
  twilio_sid  VARCHAR(100),
  error_code   VARCHAR(50),
  created_at  TIMESTAMP DEFAULT NOW(),
  updated_at  TIMESTAMP DEFAULT NOW()
);

-- Queue job lifecycle tracking
CREATE TABLE jobs_log (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  company_id  UUID REFERENCES companies(id) ON DELETE CASCADE,
  lead_id     UUID,
  rule_id     UUID,
  job_id      VARCHAR(255) NOT NULL,
  queue_name  VARCHAR(100) NOT NULL,
  status      VARCHAR(50) DEFAULT 'scheduled',
  attempts    INTEGER DEFAULT 0,
  error_msg   TEXT,
  scheduled_at TIMESTAMP,
  executed_at TIMESTAMP,
  created_at  TIMESTAMP DEFAULT NOW());

```

